Lyapunov approach to study stability of a particular class of triangular hyperbolic systems

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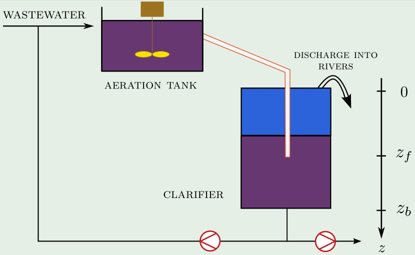
Framework



1D-Dynamic Model of the Clarifier (Two-Phase flow)

State variables: ε_s (particles density), $f_s := \varepsilon_s v_s$ (momentum) with v_s the particles velocity

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \dfrac{f_{1s}}{\rho_s}\delta_{z_f} \\ \partial_t f_s + \partial_z\left(\dfrac{f_s^2}{\varepsilon_s} + \dfrac{\sigma_e(\varepsilon_s)}{\rho_s}\right) = \underbrace{\varepsilon_s g\left(1 - \dfrac{\rho_l}{\rho_s}\right)}_{\text{Apparent weight}} + \underbrace{\dfrac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1-\varepsilon_s)}}_{\text{Friction}} + \underbrace{\dfrac{f_{2s}}{\rho_s}\delta_{z_f}}_{\text{Feed}} \end{cases}$$

Two boundary conditions (BCs) at both ends of the 1D domain are needed to close the equations, these conditions are imposed by physical devices.

$\longrightarrow$ Coupling the system with **BCs** corresponds to a *Boundary control system problem*

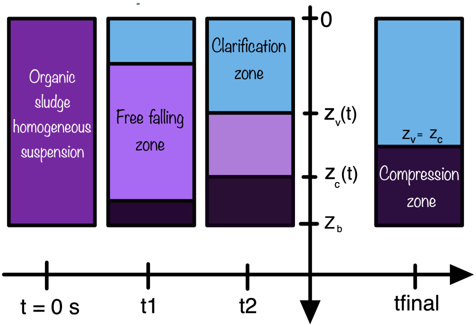
Goal 1: Design a robust boundary control that takes into account the Clarifier/Aeration Tank configuration (with disturbance inputs) in order to stabilize the sludge blanket depth

Goal 2: Prove the stability of the controlled system

Outline of Talk

Batch Column Framework

Settling in a batch column



N. Chassin, C. Valentin, J.-M. Choubert, F. Couenne, C. Jallut, 2022, *How to improve modeling of urban sludge settling in a batch column*, Note interne

C. Valentin, F. Lagoutière, J.-M. Choubert, F. Couenne, C. Jallut, 2023, *Knowledge-based model and simulations to support decision making in Wastewater Treatment*, Processes PROCEEDINGS 33rd European Symposium on Computer Aided Process Engineering (ESCAPE33), (6p), Athens. Greece

System analysis

Governing equations

$D =]0, z_b[\times]0, T[$ / Find ε_s , f_s : $D \longrightarrow \mathbb{R}$ such that :

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = 0 \\ \partial_t f_s + \partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right) = \varepsilon_s g \left(1 - \frac{\rho_l}{\rho_s}\right) - \tau \frac{\varepsilon_s^{\frac{n_r - 1}{3 - n_r}} f_s}{\rho_s (1 - \varepsilon_s)} \\ \varepsilon_s(z, 0) = \varepsilon_s^0, \quad f_s(z, 0) = f_s^0 \\ f_s(0, t) = 0, \quad f_s(z_b, t) = 0 \end{cases}$$

with : $\sigma_e(\varepsilon_s) = \begin{cases} 0 \, , & \varepsilon_s \leq \varepsilon_c \\ \sigma_0 \frac{\varepsilon_s^{n_s} - \varepsilon_c^{n_s}}{\varepsilon_c^{n_s}} \, , & \varepsilon_s > \varepsilon_c \end{cases}$

ρ_l: water density ($:= 1000 \ \frac{kg}{m^3}$)
g: gravity acceleration constant ($:= 9.81 \ \frac{m}{s^2}$)

σ_0, n_s, n_r, ε_c , τ, ρ_s (parameters)

System analysis

$$\partial_t \underbrace{\begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{U}} + \partial_z \underbrace{\begin{pmatrix} f_s \\ \frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \end{pmatrix}}_{\mathbf{F(U)}} = \underbrace{\begin{pmatrix} 0 \\ \varepsilon_s g(1 - \frac{\rho_l}{\rho_s}) - \tau \frac{\varepsilon_s^{\frac{n_r-1}{3-n_r}} f_s}{\rho_s(1-\varepsilon_s)} \end{pmatrix}}_{\mathbf{S(U)}}$$

Quasi-linear form :

$$\partial_t \mathbf{U} + \mathbf{A(U)} \cdot \partial_z \mathbf{U} = \mathbf{S(U)}$$

with $\mathbf{A(U)} = \nabla_{\mathbf{U}}\, \mathbf{F(U)} \in \mathcal{M}_2(\mathbb{R})$

$$\mathrm{Sp}(\mathbf{A}) = \{v_s - \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s-1}}\ ,\ v_s + \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s-1}}\}\ \ ;\ si\ \varepsilon_s > \varepsilon_c$$

$\Rightarrow$ Strictly hyperbolic system

$$\mathrm{Sp}(\mathbf{A}) = \{v_s\ ,\ v_s\}\ \ ;\ si\ \varepsilon_s \leq \varepsilon_c$$

$\Rightarrow$ Weakly hyperbolic system

System analysis

Map to follow

- **Search of steady-states: $\mathbf{Y}(t, z) = \mathbf{Y}^*(z), \forall t \geq 0$**
- **Linearize the system around it**
- **Find stability conditions for the linearized system**
- **Concluding stability of the steady-states for the nonlinear system may require additional conditions because of the infinite dimensional framework**

Steady-states of the system ?
Source term implies **nonuniform steady states**

Notations:

$$K = g(1 - \frac{\rho_l}{\rho_s}) \ / \ \gamma = \frac{\tau}{\rho_s} \ / \ \beta = \frac{n_r - 1}{3 - n_r}$$

System analysis

Find the steady-states: solve the following system of ODEs:

$$\begin{cases} \dfrac{\mathrm{d}f_s^*}{\mathrm{d}z} = 0, \\ \dfrac{\mathrm{d}\left(\frac{f_s^{*2}}{\varepsilon_s^*} + \frac{\sigma_e(\varepsilon_s^*)}{\rho_s}\right)}{\mathrm{d}z} = K\varepsilon_s^* - \gamma\dfrac{(\varepsilon_s^*)^\beta f_s^*}{1-\varepsilon_s^*}, \\ f_s^*(0) = 0, \quad f_s^*(z_b) = 0 \end{cases}$$

Set of steady-states

$$f_s^*(z) = 0, \quad \forall z \in [0, z_b]$$

$$\varepsilon_s^*(z) = \begin{cases} 0, & 0 \le z \le z_c \\ \left(\varepsilon_c^{n_s-1} + \frac{(n_s-1)K}{M}(z-z_c)\right)^{\frac{1}{n_s-1}}, & z_c < z \le z_b \end{cases}$$

How to determine z_c ? (to fully define the set of steady-states)

System analysis

Conservation principle of mass

$$\int_{z_c}^{z_b} \varepsilon_s^*(z)dz = \int_0^{z_b} \varepsilon_s^0(z)dz$$

For *constant initial condition*:

$$\int_{z_c}^{z_b} \varepsilon_s^*(z)dz = z_b\varepsilon_s^0$$

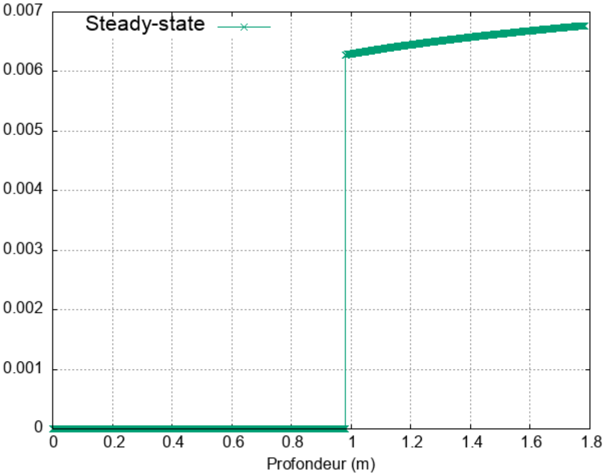
which leads to solving **for** z_c the following non-linear algebraic equation:

$$\left(\frac{M\varepsilon_c^{n_s-1}}{Kn_s} - \frac{(1-n_s)(z_b-z_c)}{n_s}\right)\left(\varepsilon_c^{n_s-1} + \frac{K(z_b-z_c)(n_s-1)}{M}\right)^{\frac{1}{n_s-1}} - \frac{M\varepsilon_c^{n_s}}{Kn_s} - z_b\varepsilon_s^0 = 0$$

Result

For any given initial condition $\mathbf{U}^0$ and fixed set of parameters (σ_0, n_s, n_r, ε_c, τ, ρ_s), we can determine analytical expressions of steady-states

Numerical simulation (Godunov-type Finite Volume scheme)



Also desired steady-state after design of boundary controller for the **continuous framework settling**

Numerical simulation (Godunov-type Finite Volume scheme)

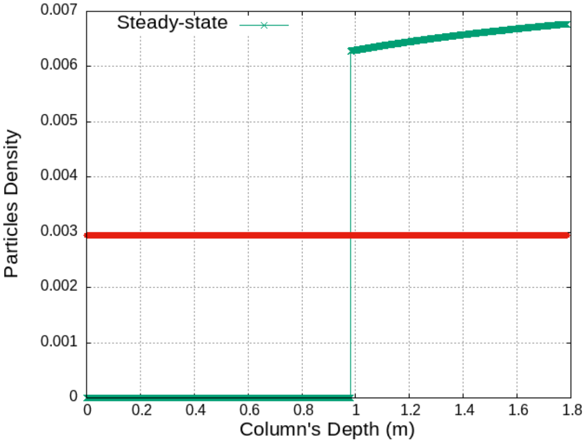


Fig. Simulation of the particles density on 6400 cells by the Godunov scheme with an LLF Riemann Solver for flux computation: comparison with steady-state

Linearization around steady-states

System in *primitive* variables:

$$\begin{cases} \partial_t \varepsilon_s + \partial_z(\varepsilon_s v_s) = 0, \\ \partial_t v_s + \partial_z(\frac{v_s^2}{2}) + \frac{1}{\rho_s \varepsilon_s}\partial_z \sigma_e(\varepsilon_s) = K - \gamma \frac{\varepsilon_s^\beta v_s}{1-\varepsilon_s} \end{cases} \tag{1}$$

For smooth solutions, system (1) is equivalent to the system in *conservative* variables, *but not for weak solutions*

Let us define the deviations of the states with respect to steady-states:

$$E_s := \varepsilon_s(t,x) - \varepsilon_s^*(x), \qquad V_s := v_s(t,x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \mathbf{Y} + \mathbf{A}^*(z)\cdot\partial_z \mathbf{Y} + \mathbf{B}^*(z)\cdot \mathbf{Y} = 0$$

$$\mathbf{Y} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma_e'(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

Linearization around steady-states

Result

When the system is strictly hyperbolic ($z_c < z \leq z_b$), Riemann Invariants are:

$$R_1^* = V_s + \frac{\sqrt{\frac{\sigma_e'(\varepsilon_s^*)}{\rho_s}}}{\varepsilon_s^*} E_s, \qquad R_2^* = V_s - \frac{\sqrt{\frac{\sigma_e'(\varepsilon_s^*)}{\rho_s}}}{\varepsilon_s^*} E_s$$

In this case, besides being linear, the system can also be put into *characteristic form*

Challenges

How to do when the system is weakly hyperbolic ? No diagonilizing transformation available
And then how to deal simultaneously with both cases (handling the strictly-weakly switch) ? Looking for a C^1 transformation that rather leads to a ~~diagonal~~ **triangular form**

Through a simplified clarifier

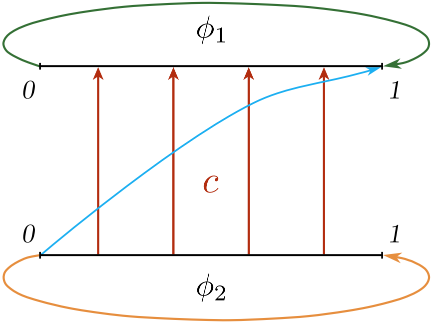
Stability analysis of a simplified clarifier: a triangular hyperbolic system

$$\begin{cases} \partial_t \phi(t,x) = \Lambda(x)\left(A(x)\partial_x \phi(t,x) + B(x)\phi(t,x)\right), \; x \in (0,1), t \geq 0 \\ \phi(t,1) = D\phi(t,0), \\ \phi(0,x) = \phi^0(x) \end{cases} \tag{2}$$

$\phi(t,x) = (\phi_1(t,x), \phi_2(t,x))$ is the state, $\Lambda(x) = \begin{bmatrix} \lambda_1(x) & 0 \\ 0 & \lambda_2(x) \end{bmatrix}$ satisfy $\lambda_{1,2}(x) > 0$ such that there exists some $x \in [0,1]$ making $\lambda_1(x) = \lambda_2(x)$, $A(x) = \begin{bmatrix} 1 & c(x) \\ 0 & 1 \end{bmatrix}$ where c is a continuous non-zero function, $B(x) = \begin{bmatrix} B_{11}(x) & B_{12}(x) \\ 0 & B_{22}(x) \end{bmatrix}$ the source terms and $D = \begin{bmatrix} D_{11} & D_{12} \\ 0 & D_{22} \end{bmatrix}$ such that $|D_{11}|, |D_{22}| < 1$.

Stability analysis of a simplified clarifier: a triangular hyperbolic system

$$\begin{cases} \partial_t \phi(t,x) = \Lambda(x)\left(A(x)\partial_x \phi(t,x) + B(x)\phi(t,x)\right), \; x \in (0,1), t \geq 0 \\ \phi(t,1) = \begin{bmatrix} D_{11} & D_{12} \\ 0 & D_{22} \end{bmatrix} \phi(t,0), \\ \phi(0,x) = \phi^0(x) \end{cases}$$



Schematic representation of the system

Stability analysis of a simplified clarifier: a triangular hyperbolic system

State space

$X = L^2((0,1);\mathbb{R}) \times H^1((0,1);\mathbb{R})$ equipped with the norm:

$$\|\cdot\|_X^2 : (f,g) \mapsto \|f\|^2_{L^2((0,1),\mathbb{R})} + \|g\|^2_{L^2((0,1);\mathbb{R})} + \|g'\|^2_{L^2((0,1);\mathbb{R})}$$

Exponential stability

The origin of system (2) is δ-exponentially stable if there exists an overshoot $\kappa \geq 1$ and a decay rate $\delta > 0$ such that for any initial condition $\phi^0 \in X$:

$$\|\phi(t;\phi^0)\|_X^2 \leq \kappa e^{-\delta t}\|\phi^0\|_X^2, \ \forall t \geq 0 \tag{3}$$

Sufficient conditions that ensures exponential stability of this system ?

To answer, stability analysis is performed. The study will be divided into two cases: Without and with source term B

First case: B = 0

Assumption 1

The function λ_2 and the matrix D satisfy the following inequality:

$$\max(D_{11}^2, D_{22}^2) < 1 - \frac{\overline{\lambda_2'}}{\underline{\lambda_2}}.$$

Theorem 1

Let Assumption 1 holds and let $\nu := \frac{\overline{\lambda_2'}}{\underline{\lambda_2} - \overline{\lambda_2'}}$, $\bar{\nu} := \frac{1 - D_{22}^2}{D_{22}^2}$. Then, for any $\delta < \min\{\frac{\bar{\nu}\underline{\lambda_1}}{1+\bar{\nu}}, \frac{\bar{\nu}\underline{\lambda_2}}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\underline{\lambda_2}}{(1+\bar{\nu})(1+\nu)}\}$, the origin of system (2) is δ-exponentially stable.

First case: B = 0

Proof: The method used to prove Theorem 1 relies on the construction of a Lyapunov functional
Consider the following candidate functional:

$$\mathcal{V}(\phi) = \mathcal{V}_1(\phi) + \alpha_{20}\mathcal{V}_{20}(\phi) + \alpha_{21}\mathcal{V}_{21}(\phi)$$

With

$$\begin{aligned} \mathcal{V}_1(\phi) &= \int_0^1 \tfrac{1+\mu x}{\lambda_1(x)}\phi_1^2(x)dx, \\ \mathcal{V}_{20}(\phi) &= \int_0^1 \tfrac{1+\mu x}{\lambda_2(x)}\phi_2^2(x)dx, \\ \mathcal{V}_{21}(\phi) &= \int_0^1 \tfrac{1+\mu x}{\lambda_2(x)}(\partial_x\phi_2)^2(x)dx \end{aligned}$$

and scalars $\mu, \alpha_{20}, \alpha_{21} > 0$.
Recall that $\mathcal{V}$ induces a norm on X that is equivalent to the state norm $\|\cdot\|_X$

First case: B = 0

Time derivative of each term of $\mathcal{V}$ along C^1 solutions of system (2)

$$\begin{aligned} \dot{\mathcal{V}}_{20}(\phi) &= 2\int_0^1 (1+\mu x)\phi_2(x)\partial_x\phi_2(x)dx, \\ &= -\mu\int_0^1 \phi_2^2(x)dx + (1+\mu)\phi_2^2(1) - \phi_2^2(0) \end{aligned} \tag{4}$$

Second:

$$\begin{aligned} \dot{\mathcal{V}}_1(\phi) = -\mu\int_0^1 \phi_1^2(x)dx + (1+\mu)\phi_1^2(1) - \phi_1^2(0) \\ + 2\int_0^1 c(x)(1+\mu x)\phi_1(x)\partial_x\phi_2(x)dx \end{aligned} \tag{5}$$

Third:

$$\begin{aligned} \dot{\mathcal{V}}_{21}(\phi) &= -\mu\int_0^1 (\partial_x\phi_2)^2(x)dx + (1+\mu)(\partial_x\phi_2)^2(1) - (\partial_x\phi_2)^2(0) \\ &\quad + 2\int_0^1 (1+\mu x)\frac{\lambda_2'(x)}{\lambda_2(x)}(\partial_x\phi_2)^2(x)dx \end{aligned} \tag{6}$$

First case: B = 0

From (4), (5), (6) + **Young's inequality** and $\phi_2(1) = D_{22}\phi_2(0)$ we get:

$$\begin{aligned}\dot{\mathcal{V}}(\phi) \quad \leq \quad & \left(-\mu + (1+\mu)\bar{c}\gamma\right)\|\phi_1\|^2_{L^2} - \alpha_{20}\mu\|\phi_2\|^2_{L^2} \\ & + \left(\frac{(1+\mu)\bar{c}}{\gamma} + \alpha_{21}\frac{\nu(2+\mu)-\mu}{1+\nu}\right)\|\partial_x\phi_2\|^2_{L^2} \\ & + \left(\alpha_{20}(\mu-\bar{\nu}) + \left(\frac{1+\mu}{1-(1+\mu)D^2_{11}}\right)\frac{D^2_{12}}{D^2_{22}}\right)\phi_2^2(1) \\ & + \alpha_{21}(\mu-\bar{\nu})(\partial_x\phi_2)^2(1)\end{aligned}$$

By introducing $\varepsilon > 0$, and selecting the positive scalars $\mu, \alpha_{20}, \alpha_{21}, \gamma$ as follows

$$\mu = \mathcal{O}(\bar{\nu}),\ \gamma = \mathcal{O}(\varepsilon),\ \alpha_{20} = \mathcal{O}(\frac{1}{\varepsilon}),\ \alpha_{21} = \mathcal{O}(\frac{1}{\varepsilon^2}), \quad \text{for } \varepsilon \in \mathscr{O}(0)$$

We end up with:

$$\dot{\mathcal{V}}(\phi) \leq -\delta\mathcal{V}(\phi), \quad \forall\delta < \min\left\{\frac{\bar{\nu}\underline{\lambda}_1}{1+\bar{\nu}}, \frac{\bar{\nu}\underline{\lambda}_2}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\underline{\lambda}_2}{(1+\bar{\nu})(1+\nu)}\right\}$$

$$\Rightarrow \mathcal{V}(\phi(t;\phi_0)) \leq e^{-\delta t}\mathcal{V}(\phi^0)\ ,\ \forall\phi^0 \in X$$

Using the equivalence of state-space norm and Lyapunov induced norm on X,

$$\left\|\phi(t;\phi^0)\right\|^2_X \leq \kappa e^{-\delta t}\left\|\phi^0\right\|^2_X$$

Second case: B $\neq 0$

Assumption 2

The functions λ_2, B and the matrix D satisfy:

$$\max(D_{11}^2, D_{22}^2) < 1 - \max\left\{2\overline{B_{11}}, \frac{\overline{\lambda_2'}}{\underline{\lambda_2}} + 2\overline{B_{22}}\right\}$$

Theorem 2

Let Assumption (2) holds and let

$$\nu_1 := 2\overline{B_{11}}(1 - 2\overline{B_{11}})^{-1},\ \nu_2 := \frac{\overline{\lambda_2'} + 2\underline{\lambda_2}\overline{B_{22}}}{\underline{\lambda_2} - \overline{\lambda_2'} - 2\underline{\lambda_2}\overline{B_{22}}},\ \bar{\nu} := \frac{1 - D_{22}^2}{D_{22}^2}.$$

Then, for any $\delta < \min\{\delta_1, \delta_2\}$ where $\delta_1 = \frac{\underline{\lambda_1}(\bar{\nu}-\nu_1)}{(1+\nu_1)(1+\bar{\nu})}$ and $\delta_2 = \frac{\underline{\lambda_2}(\bar{\nu}-\nu_2)}{(1+\nu_2)(1+\bar{\nu})}$, the origin of system (2) is δ-exponentially stable.

Second Case: B $\neq 0$

Sketch of the Proof: Consider the same candidate Lyapunov functional $\mathcal{V}$ as before, time derivative of each additional terms along C^1 trajectories of the system are:

$$\mathcal{W}_{20}(\phi) = 2\int_0^1 (1+\mu x) B_{22}(x) \phi_2^2(x) dx,$$

$$\mathcal{W}_1(\phi) = 2\int_0^1 (1+\mu x)\left(B_{11}(x)\phi_1^2(x) + B_{12}(x)\phi_1(x)\phi_2(x)\right) dx.$$

$$\mathcal{W}_{21} = 2\int_0^1 (1+\mu x) B_{22}(x) \partial_x \phi_2^2(x) dx + 2\int_0^1 (1+\mu x)\frac{(\lambda_2 B_{22})'(x)}{\lambda_2(x)} \partial_x \phi_2(x)\phi_2(x) dx.$$

Similarly to the first case, making use of Young's inequality and selecting the appropriate parameters, we achieve:

$$\dot{\mathcal{V}}(\phi) \leq -\delta \mathcal{V}(\phi), \quad \forall \delta < \min\left(\frac{\underline{\lambda}_1(\bar{\nu}-\nu_1)}{(1+\nu_1)(1+\bar{\nu})}, \frac{\underline{\lambda}_2(\bar{\nu}-\nu_2)}{(1+\nu_2)(1+\bar{\nu})}\right)$$

and the equivalence of norms ends the proof. $\blacksquare$

Numerical results

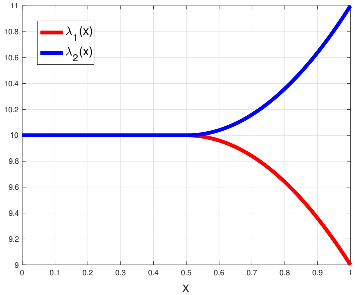
Academic Example

Consider system (2) with:

$$\lambda_1(x) = \begin{cases} 10, & x < \frac{1}{2} \\ -4(x-\frac{1}{2})^2 + 10 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$\lambda_2(x) = \begin{cases} 10, & x < \frac{1}{2} \\ 4(x-\frac{1}{2})^2 + 10 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$c(x) = x + 4,\ D = \begin{bmatrix} \frac{1}{2} & 1 \\ 0 & \frac{1}{2} \end{bmatrix},\ B = \begin{bmatrix} \frac{x}{6} & c(x) \\ 0 & \frac{x}{6} \end{bmatrix},\ \phi_1^0(x) = 0,\ \phi_2^0(x) = \frac{x^2}{2}(1 - \frac{x}{3}) - \frac{2}{3}.$$



Academic Example

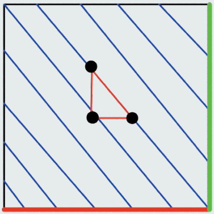
Regarding Theorem 1, the assumption $\max(D_{11}^2, D_{22}^2) = \frac{1}{4} < 1 - \frac{\overline{\lambda_2'}}{\underline{\lambda_2}} = \frac{3}{5}$ is satisfied.

Regarding Theorem 2, the assumption $\max(D_{11}^2, D_{22}^2) = \frac{1}{4} < 1 - \max\left\{2\overline{B_{11}}, \frac{\overline{\lambda_2'}}{\underline{\lambda_2}} + 2\overline{B_{22}}\right\} = \frac{4}{15}$ is satisfied.

The system is discretized using an *Upwind* type explicit finite difference scheme:

$$\begin{cases} \frac{\phi_{1,i}^{n+1} - \phi_{1,i}^n}{\Delta t} = \lambda_{1,i} \frac{\phi_{1,i+1}^n - \phi_{1,i}^n}{\Delta x} + c_i \lambda_{2,i} \frac{\phi_{2,i+1}^n - \phi_{2,i}^n}{\Delta x} \\ \frac{\phi_{2,i}^{n+1} - \phi_{2,i}^n}{\Delta t} = \lambda_{2,i} \frac{\phi_{2,i+1}^n - \phi_{2,i}^n}{\Delta x} \end{cases}$$

Characteristic lines through (x,t) plane



Setting:

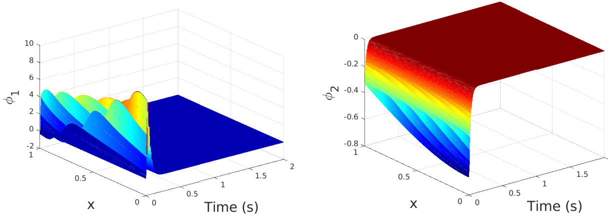
$$\gamma_{1,i} = \frac{\lambda_{1,i}\Delta t}{\Delta x}, \quad \gamma_{2,i} = \frac{c_i \lambda_{2,i}\Delta t}{\Delta x}$$

finally we get the numerical scheme:

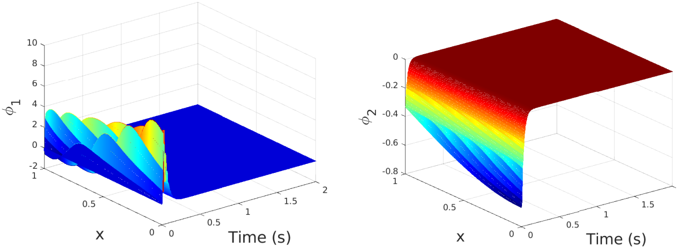
$$\phi_{1,i}^{n+1} = (1 - \gamma_{1,i})\phi_{1,i}^n + \gamma_{1,i}\phi_{1,i+1}^n - \gamma_{2,i}\phi_{2,i}^n + \gamma_{2,i}\phi_{2,i+1}^n$$

$$\phi_{2,i}^{n+1} = (1 - \gamma_{2,i})\phi_{2,i}^n + \gamma_{2,i}\phi_{2,i+1}^n$$

Academic example: Simulation



System (2) without source term: (left) state ϕ_1 / (right) state ϕ_2



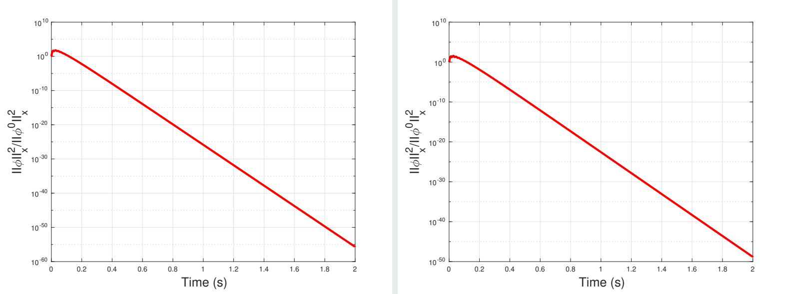
System (2) with source term: (left) state ϕ_1 / (right) state ϕ_2

Academic example: Decay Rate

Recall δ-exponential stability:

$$\|\phi(t;\phi^0)\|_X^2 \leq \kappa e^{-\delta_{th} t}\|\phi^0\|_X^2,\ \forall t \geq 0$$

$$\Rightarrow \ln\left(\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}\right) \leq \ln(\kappa) - \delta_{th} t$$



$\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}$ **in log scale: (left) without source term / (right) with source term**

Academic example: Decay Rate

Without source term, theoretically, decay rate is given by $\min\{\frac{\bar{\nu}\underline{\lambda}_1}{1+\bar{\nu}}, \frac{\bar{\nu}\underline{\lambda}_2}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\underline{\lambda}_2}{(1+\bar{\nu})(1+\nu)}\} = \frac{7}{2}$, whereas, numerically, decay rate is

$$\delta_{num} = 68.50 > \delta_{th}$$

With source term, decay rate is at least equal to $\delta_{th} = 0.16$, whereas, numerically, decay rate is:

$$\delta_{num} = 60.19 > \delta_{th}$$

The difference is due to the conservatism introduced by the proposed Lyapunov analysis

Conclusions and perspectives

Paper submission for 4th IFAC conference of MICNON

M.O. Amirat, V. Andrieu, J. Auriol, M. Bajodek and C. Valentin, *Stability analyis of a class of 2 x 2 triangular hyperbolic systems*

Ongoing work

Tackle the case where the system **is not in cascade** form $\Rightarrow$ Choose a wider class of controllers (of Dynamical type ?)
Conclude about the stability of the nonlinear system
Apply to the Clarifier/Aeration tank system.

Thank you for your attention.
Questions ?